Matias Shedden

Curriculum Vitae

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Education

August 2022–May 2027 **Ph.D in Statistics**, University of Florida, Gainesville, United States

August 2018–May 2022 **B.S. in Applied Mathematics, B.S. in Chemistry**, GPA 4.02/4.33, University of New Mexico, Albuquerque, United States

Research Interests

- Near-Boundary Asymptotic Theory, Adaptive Clinical Trial Design, Statistical Computing, Linear Mixed Models, and Time Series.

Papers

- **M. Shedden** and K. O. Ekvall (2026+). Score-based confidence intervals for variance-covariance parameters in linear mixed models. (Submitted).
- **M. Shedden** and K. O. Ekvall (2026+). Near-boundary asymptotics for profile score statistics. (Paper in progress).

Presentations

- Poster: "Reliable score-based confidence intervals for covariance parameters in linear mixed models." *Southern Regional Council on Statistics Summer Research Conference*, June 2026. (**Best Poster Award Winner**)
- Seminar (*UF Student Seminar Series*): "Uniform Inference near the boundary." *University of Florida*, November 2025.
- Poster: "Machine-learning-based subgroup identification in an adaptive clinical trial." *Conference for Statistical Foundations of AI (CSFAI) at UCSB*, November 2026 (upcoming).
- Poster: "Machine-learning-based subgroup identification in an adaptive clinical trial." *Academy of Science, Engineering, and Medicine of Florida Annual Meeting.*, November 2026 (upcoming).

Research Positions

Summer 2026 **Graduate Research Intern**, Department of Biostatistics and Bioinformatics, Moffitt Cancer Center. Advisor: Dr. Vivien Yin

- Research area: Adaptive clinical trial design using causal forests. Part of dissertation work, paper in progress.

Teaching

- STA 3100: Programming with Data in R (Primary Instructor), Summer A 2023, Spring 2024, and Fall 2026.
- STA 3032: Engineer Statistics (Primary Instructor), Fall 2023 and Fall 2025.
- STA 4322: Introduction to Statistics Theory (Primary Instructor), Fall 2024 and Spring 2026.
- STA 3100: Programming with Data in R (Teaching Assistant), Fall 2022 and Spring 2023

Honors & Awards

- M. Clinton Miller, III Award for Best Student Poster at the Southern Regional Council on Statistics Summer Research Conference, 2026.
- Undergraduate Honors Program: Undergraduate thesis, "Asymptotic Waveform Evaluation of Gravitational Waves", advised by Dr. Stephen Lau. Graduated with honors (magna cum laude) in mathematics.
- UNM Arts and Sciences Support for Undergraduate Research Experience (ASSURE) Fellowship, Fall 2021.

Other Research Experience

June–August 2019 **Laboratory Assistant**, UNM Department of Chemistry, under Dr. John Grey

June 2020–May 2022 **Research Technician (Data Analyst)**, UNM College of Pharmacy, Department of Pharmaceutical Science, under Dr. Esther Erdei.

April 2021–May 2022 **Undergraduate Researcher**, UNM Department of Mathematics and Statistics, under Dr. Stephen Lau.

Other Papers

- E. Erdei, D. Torgerson, R. O'Leary, M. Spear, M. Shedden, M. O'Leary, K. Enright, L. Best. "Respiratory syncytial virus-specific immunoglobulin G (IgG) concentrations associate with environmental and genetic factors: the Factors Influencing Pediatric Asthma Study."

Other Work

- Book: "Engineering Statistics", Matias Shedden, 2023. 201-Page book for teaching STA 3032 at the University of Florida, available on my website.
- Course Material for STA 3100. Designed the curriculum for the class, including lectures, assignments, quizzes, and exams. This material was used by other graduate students to teach the same class, at the request of the department.